

Avionics & Mechanisms

AVDASI 2 · Week 1 introduction

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Today

Opening: who we are, where things happen, and why this matters to every division

Part 1: Avionics

The requirements, integration, the workshops, and what to do before Friday

Part 2: Mechanisms

The requirements, integration, and the sessions

Close: where to find things, and questions

Who's who

Teaching

- **Steve Bullock:** avionics lead
- **George:** teaching and coordination, Thu & Fri
- **Robin Carter:** PhD from 1 Oct, has flown these autopilots on the MSc Aerial Robotics project

Labs and kit

- **Tim:** avionics kit issue and return, M.003
- **Jasmin:** lab coverage

Unit coordinators

- **Jasmin and Mark**

Where things happen

Space	Room	What for
Stacks Room	QB F.05, under the QB library	Avionics workshops, Fri 09:00-11:00
Avionics lab	M.003	Kit issue, supervised tool use
The Hangar	L.080	Main lab, open access
Fabrication lab	O.080	Fabrication

Why avionics and mechanisms?

Every division depends on them



The model is only as good as what moves it

- Your wind-tunnel model has **control surfaces**, and something has to move them repeatably
- Your test campaign needs **data**, and something has to measure, time-stamp and log it
- Your aerodynamicists need **the angle you asked for**, not the one you got

Avionics and mechanisms turn a shaped piece of foam and carbon into an **instrumented test article**.

Part 1: Avionics

Flight-grade kit, off-design



Flight-grade kit, off-design

- **Cube Orange+** autopilot running **ArduPilot**, the Flight Lab's standard research autopilot
- Flown for volcano monitoring, conservation, humanitarian work and search and rescue
- In AVDASI 2 it's **bolted into a wind tunnel**, with no GPS and restricted movement, and you command surfaces directly

That's deliberately off-design. Much of the avionics work is getting a flight controller to behave usefully on the ground.

Avionics requirements

Placeholder. Fill in from the requirements specification: the avionics requirements, with their IDs.

- requirement: what it means in practice
- requirement: what it means in practice
- requirement: what it means in practice

What "done" looks like

Placeholder. The avionics minimum working example (MWE), stated as the stream's own internal requirements, plus how it's assessed.

- The step-by-step guide gets every avionics team to a **minimum working example**
- The MWE has its own short requirements: a deliverable for the avionics stream, separate from the unit specification
- Your design builds on it; there are many routes to the requirements

Avionics is an integration job

With	You need to agree
Aerodynamics	Surface sizes and deflections, and so hinge moments, and so servo torque
Mechanisms	Linkages, travel, backlash, end stops
Structures	Mounting, access, cable routes, mass
Test	Power, comms, what's logged, how it's handed over

Every row is an **interface**. Interfaces agreed late are the ones that fail.

Interfaces to pin down early

- **Torque and travel:** what the surface needs, and what the servo can give with margin at a stall
- **Power:** what each component draws, at what voltage, from which supply
- **Connectors and cables:** which plug, which route, and who crimps it
- **Data:** what's logged, at what rate, in what units and file format, and who uses it
- **Command:** who moves which surface in the tunnel, and how you stop it

The system on the bench and in the tunnel

Power

- Bench supply → power module → Cube
- Bench supply → BEC → servo rail → servos
- *Not* laptop USB for anything that moves

Signals

- Cube → servo signal pins → surfaces
- Sensors → ADC / I²C → Cube → log
- Cube ↔ Wi-Fi telemetry ↔ laptop running Mission Planner

Avionics workshops

Week	Date	Where	Provisional focus
1	Fri 25 Sep, 09:00-11:00	Stacks Room + M.003	Kit issue; Cube, Mission Planner, telemetry, power
2	Fri 2 Oct, 09:00-11:00	Stacks Room	Servos and Lua scripting (run by George)
3	No session		Company Day, Fri 9 Oct
4	Fri 16 Oct, 09:00-11:00	Stacks Room	Sensors: ADC and I ² C
5	Fri 23 Oct, 09:00-11:00	Stacks Room	To be confirmed: extra support

Friday: kit day

1. **09:00** A short taught intro in the Stacks Room: the kit, and the requirements in more depth
2. **From 09:30** Groups go up to **M.003** one at a time to collect a kit from Tim
3. **While you wait** Read the guide's first pages (*Kit*, *Cube*), with no hardware yet
4. **Kit in hand** Work through the guide to the Friday checkpoint
5. **To finish** How not to break each component

Friday checkpoint: Cube on Plane firmware, buzzer quiet, connected over Wi-Fi, running on bench power.

Before Friday

- **Install Mission Planner on a Windows laptop.** It's Windows-only; one working laptop per group at minimum
- **Bring** that laptop, charged
- **Read** the guide's index, *Kit* and *Cube* pages

avdasi2.github.io → Avionics → Step-by-step guide

Part 2: Mechanisms

From servo to surface



What mechanisms do

Placeholder. Plan in the mechanisms strand review.

- Turn a servo's rotation into the surface deflection that the test needs
- Hold that deflection repeatably, under aerodynamic load
- Survive installation, adjustment and a wind-tunnel campaign

Mechanisms requirements

Placeholder. Fill in from the requirements specification: the mechanisms requirements, with their IDs.

- requirement: what it means in practice
- requirement: what it means in practice

Mechanisms integration

With	You need to agree
Avionics	Servo choice, horn and mounting; travel, end stops, the neutral position
Aerodynamics	Deflection range and accuracy; hinge moments
Structures	Hinge lines, load paths, access for adjustment
Test	How deflection is set and checked in the tunnel

Mechanisms sessions

Week	When	Where	Focus
3	Thu 8 Oct, 11:00-13:00	1.59 Design Suite	To be confirmed
4	Thu 15 Oct, 11:00-13:00	1.59 Design Suite	To be confirmed
5	Thu 22 Oct, 11:00-13:00	1.59 Design Suite	To be confirmed

Where to find things

- **The guide:** avdasi2.github.io
- **Unit information, spec and assessment:** SharePoint
- **Questions:** Teams, or ask in the workshops
- **Found an error in the guide?** Raise an issue or a pull request on GitHub; it's open for exactly that

Questions

See you on Friday, 09:00, Stacks Room

